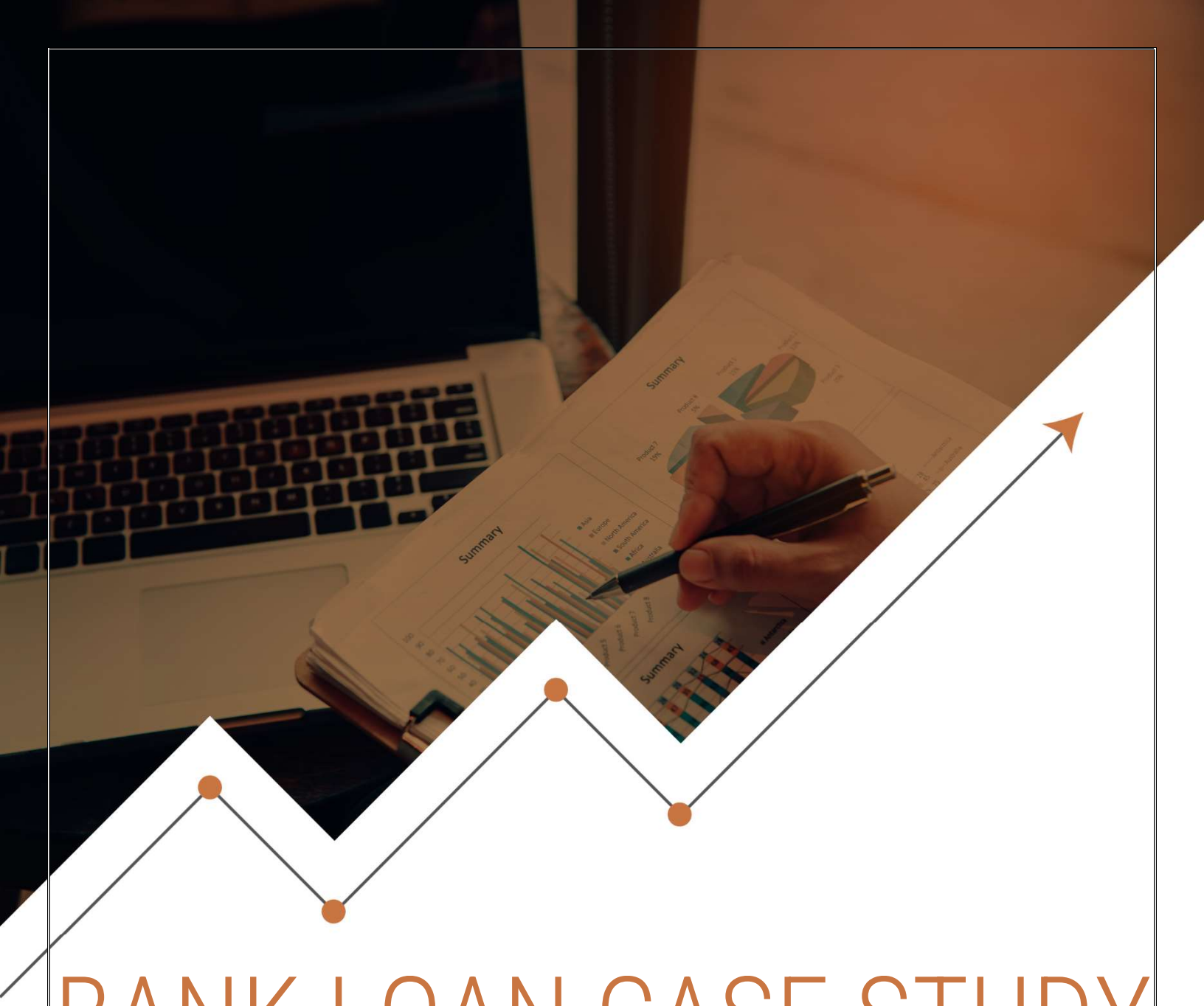


BANK LOAN CASE STUDY

(Final Project – 2)



wordexceltemplates.com

Bank Loan Case Study

Final Project:2

::

Project Description

Problem Statement - you are a data analyst at a finance company that specializes in lending various types of loans to urban customers. Your company faces a challenge: some customers who do not have a sufficient credit history take advantage of this and default on their loans. Your task is to use Exploratory Data Analysis (EDA) to analyze patterns in the data and ensure that capable applicants are not rejected.

By doing this project we will be able to do analysis for Bank loan.

Downloaded the Dataset for this project and the dataset contains various files like

- **Application Data -** Provides details about the current loan applications.
- **Previous Data -** Contains information about previous loan applications.
- **Column Description -** Describes the columns present in the other datasets, explaining what each column represents.
- **Important Notes for this project**

Tools Used –

- MS Excel (office365) for Analysis.
- MS Word (Office365) for Detailed report.

Data Analytics Tasks:

A. Identify Missing Data and Deal with it Appropriately: As a data analyst, you come across missing data in the loan application dataset. It is essential to handle missing data effectively to ensure the accuracy of the analysis.

Task: Identify the missing data in the dataset and decide on an appropriate method to deal with it using Excel built-in functions and features.

Action Taken: --

For Identifying Missing Data –

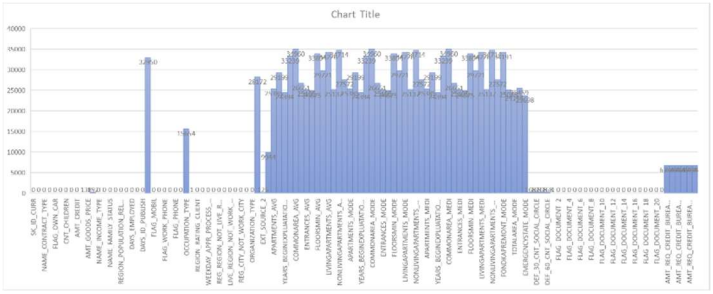
- First of all, counted No of cells using (CountA) Function then counted Blank cells using formula **(=COUNTBLANK(A5:A50003)).**
- Then checked for Duplicate values, no Duplicate values found.
- Deleted Columns having large no of Missing data.
- Filled the data for remaining columns using mean/median function.
- Formatted the file is Table format.

File Details-

1. **Application_data –**

- Before Formatting –

  - **Rows -** 49999
  - **Columns -** 122
  - **Count blank cells –** 1488212.



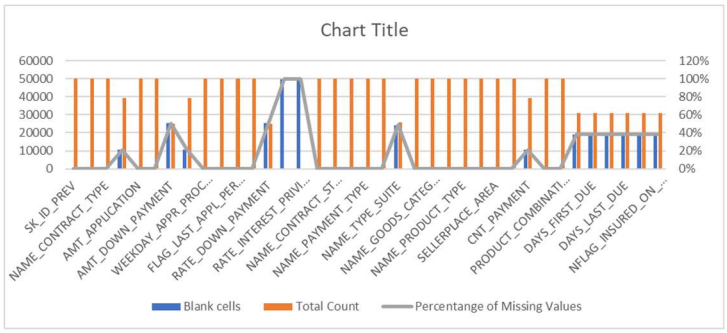
- File after Formatting –

  - **Rows -** 49999
  - **Columns -** 42
  - **Blank cells –** 0.

2. **Previous_application –**

- File Before Formatting –

  - **Rows –** 49999
  - **Column –** 37
  - **Blank cells –** 321203



- File after Formatting –

    - **Rows -** 49999
    - **Columns -** 26
    - **Blank cells –** 0.

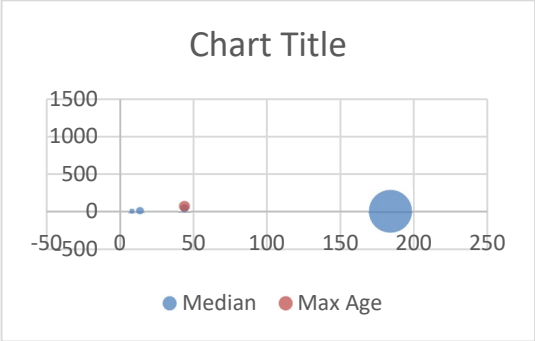
B. Identify Outliers in the Dataset: Outliers can significantly impact the analysis and distort the results. You need to identify outliers in the loan application dataset.

> **Task:** Detect and identify outliers in the dataset using Excel statistical functions and features, focusing on numerical variables.

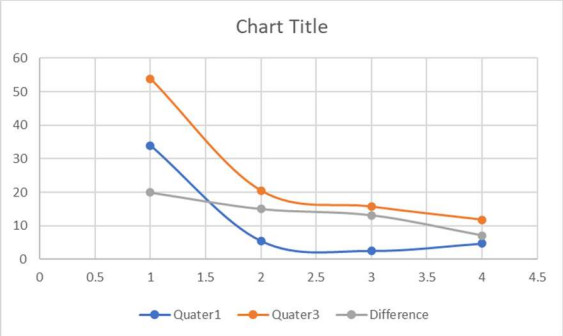
Action Taken: -

Calculated functions and created chart -

Functions	Outliners(Age)	Outliners(DAYS_REGISTRATION(Years))	Outliners(DAYS_EMPLOYED(Years))	Outliners(DAYS_ID_PUBLISH(Years))
Mean	44	14	184	8
Median	43	12	6	9
Std Deviation	11.94892234	9.65893986	380.7027603	4.134850183
Max Age	69	61	1001	17
Min Age	21	0	0	0



Quartile				
Functions	Outliners(Age)	Outliners(DAYS_REGISTRATION(Years))	Outliners(DAYS_EMPLOYED(Years))	Outliners(DAYS_ID_PUBLISH(Year))
Quater1	33.91369863	5.473972603	2.556164384	4.717808219
Quater3	53.81917808	20.44794521	15.66575342	11.77260274
Difference	19.90547945	14.9739726	13.10958904	7.054794521



C. Analyze Data Imbalance: Data imbalance can affect the accuracy of the analysis, especially for binary classification problems. Understanding the data distribution is crucial for building reliable models.

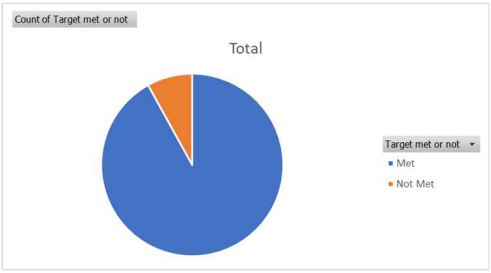
Task: Determine if there is data imbalance in the loan application dataset and calculate the ratio of data imbalance using Excel functions.

Action Taken: -

Counted the Target status using Countif function.

Target	Count	Formula
Met(0)	45973	=COUNTIF(B2:B50000,"0")
Not Met(1)	4025	=COUNTIF(B3:B50001,"1")

Target Status	Count of Target Status
Met	45973
Not Met	4026
Grand Total	**49999**



D. Perform Univariate, Segmented Univariate, and Bivariate Analysis: To gain insights into the driving factors of loan default, it is important to conduct various analyses on consumer and loan attributes.

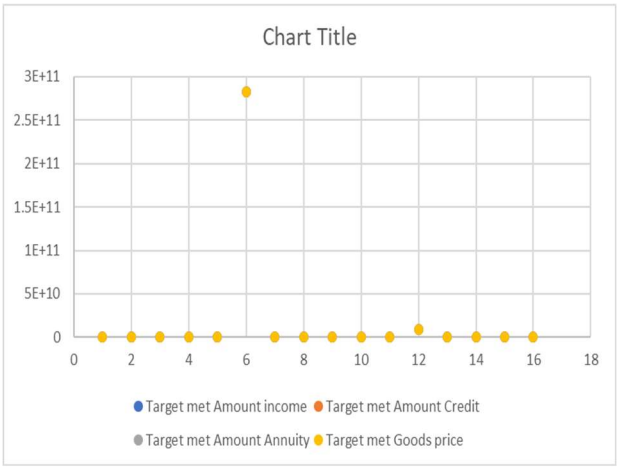
Task: Perform univariate analysis to understand the distribution of individual variables, segmented univariate analysis to compare variable distributions for different scenarios, and bivariate analysis to explore relationships between variables and the target variable using Excel functions and features.

Action Taken: -

Did Descriptive Statics using data analysis tool in excel.

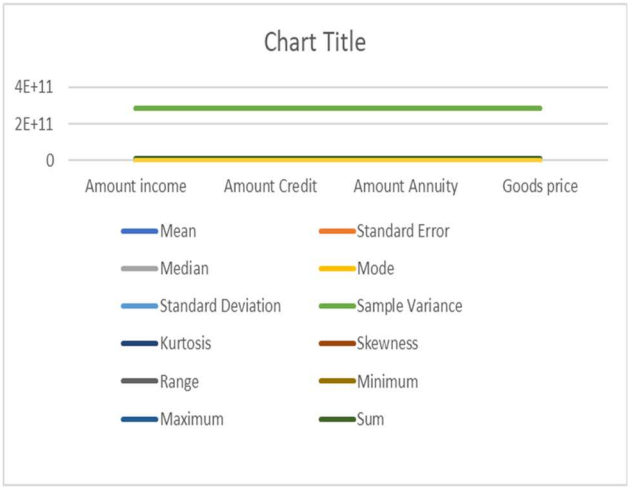
Target met Descriptive statistics.

Function	Amount income	Amount Credit	Amount Annuity	Goods price
Mean	170766.9558	170766.9558	170766.9558	170766.9558
Standard Error	2378.438566	2378.438566	2378.438566	2378.438566
Median	145800	145800	145800	145800
Mode	135000	135000	135000	135000
Standard Deviation	531824.3947	531824.3947	531824.3947	531824.3947
Sample Variance	2.82837E+11	2.82837E+11	2.82837E+11	2.82837E+11
Kurtosis	46581.60183	46581.60183	46581.60183	46581.60183
Skewness	212.0757023	212.0757023	212.0757023	212.0757023
Range	116974350	116974350	116974350	116974350
Minimum	25650	25650	25650	25650
Maximum	117000000	117000000	117000000	117000000
Sum	8538006258	8538006258	8538006258	8538006258
Count	49998	49998	49998	49998
Largest(1)	117000000	117000000	117000000	117000000
Smallest(1)	25650	25650	25650	25650
Confidence Level(95.0%)	4661.766785	4661.766785	4661.766785	4661.766785



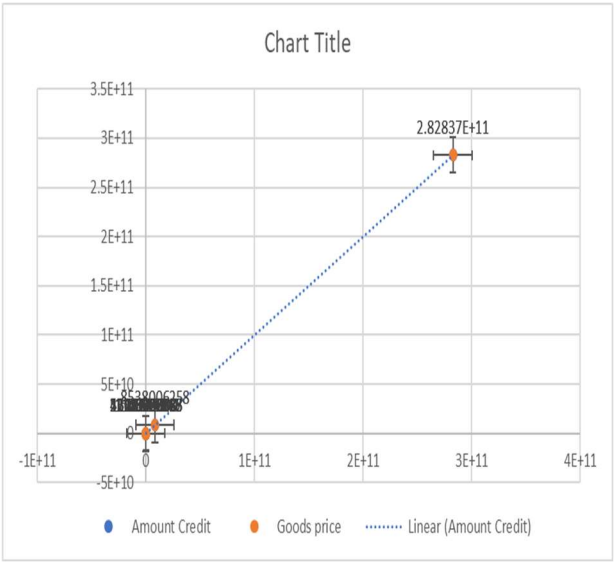
Target not met Descriptive statistics.

Target Not met				
Function	Amount income	Amount Credit	Amount Annuity	Goods price
Mean	170766.9558	170766.9558	170766.9558	170766.9558
Standard Error	2378.438566	2378.438566	2378.438566	2378.438566
Median	145800	145800	145800	145800
Mode	135000	135000	135000	135000
Standard Deviation	531824.3947	531824.3947	531824.3947	531824.3947
Sample Variance	2.82837E+11	2.82837E+11	2.82837E+11	2.82837E+11
Kurtosis	46581.60183	46581.60183	46581.60183	46581.60183
Skewness	212.0757023	212.0757023	212.0757023	212.0757023
Range	116974350	116974350	116974350	116974350
Minimum	25650	25650	25650	25650
Maximum	117000000	117000000	117000000	117000000
Sum	8538006258	8538006258	8538006258	8538006258
Count	49998	49998	49998	49998
Largest(1)	117000000	117000000	117000000	117000000
Smallest(1)	25650	25650	25650	25650
Confidence Level(95.0%)	4661.766785	4661.766785	4661.766785	4661.766785



Both Descriptive statistics (met + not met).

All Stat				
Function	Amount income	Amount Credit	Amount Annuity	Goods price
Mean	170766.9558	170766.9558	170766.9558	170766.956
Standard	2378.438566	2378.438566	2378.438566	2378.43857
Median	145800	145800	145800	145800
Mode	135000	135000	135000	135000
Standard	531824.3947	531824.3947	531824.3947	531824.395
Sample '	2.82837E+11	2.82837E+11	2.82837E+11	2.8284E+11
Kurtosis	46581.60183	46581.60183	46581.60183	46581.6018
Skewnes	212.0757023	212.0757023	212.0757023	212.075702
Range	116974350	116974350	116974350	116974350
Minimum	25650	25650	25650	25650
Maximun	117000000	117000000	117000000	117000000
Sum	8538006258	8538006258	8538006258	8538006258
Count	49998	49998	49998	49998
Largest(1	117000000	117000000	117000000	117000000
Smallest(	25650	25650	25650	25650
Confiden	4661.766785	4661.766785	4661.766785	4661.76679

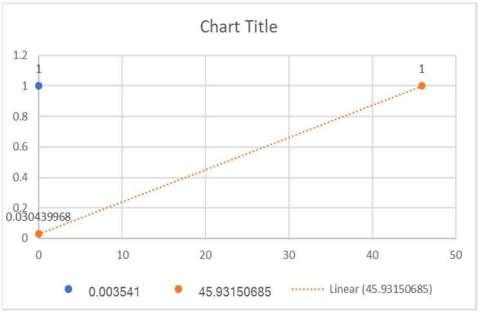


E. Identify Top Correlations for Different Scenarios: Understanding the correlation between variables and the target variable can provide insights into strong indicators of loan default.

Task: Segment the dataset based on different scenarios (e.g., clients with payment difficulties and all other cases) and identify the top correlations for each segmented data using Excel functions.

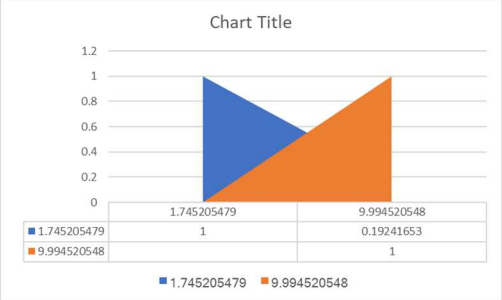
Target Met – made correlation using Data analysis Tool in Excel.

	0.00354	*45.9315*
0.003541	1	
45.93151	0.03044	1



Target not met - made correlation using Data analysis Tool in Excel.

	1.74521	*9.99452*
1.745205	1	
9.994521	0.192417	1



File link-

https://docs.google.com/spreadsheets/d/1mS8UEKHqadCC7uEx8uB1v2iMNb_hMT2m/edit?usp=drive_link&ouid=107355046625792303579&rtpof=true&sd=true

https://docs.google.com/spreadsheets/d/1HOR26bhaXNlGJmmIqp9GYlNMQZ916Jep/edit?usp=drive_link&ouid=107355046625792303579&rtpof=true&sd=true

~~END~~